The Application of DSRC Technology in Intelligent Transportation System

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Abstract

DSRC (Dedicated Short Range Communications) technology is specially developed for vehicle-to-vehicle and vehicle-to-infrastructure communication in Intelligent Transportation System (ITS). Compared to other wireless communication technologies, DSRC technology has many advantages. This paper discusses the technical characteristics, communication mechanism, and the application of DSRC technology in ITS.

1 Introduction

DSRC (Dedicated Short Range Communications) is one of the microwave communication technologies based on long distance RFID. The transport information and control system of International Organization for Standardization (ISO/TC 2004) is in charge of making the international standard of DSRC. There are three DSRC standard camps in the world: Europe ENV Series, USA 900 MHz and Japan ARIBSTD-T75 Standard. [1] ITS Center affiliated to China's Ministry of Transportation has advised Wireless Management Congress to allocate 5.8GHz frequency range. With the application of wireless technology like DSRC, vehicles will be able to communicate with each other in order to improve driving experience and enhance traffic safety.

2 The technical characteristics of DSRC

DSRC technology can supply the high speed wireless communication service to vehicle-to-infrastructure (V2I) communication, vehicle-to-vehicle (V2V) communication and other transportation service. DSRC has a high data transfer rate and small communication delay. It is widely used in ETC (Electronic Toll Collection) and public safety. It can build a high speed and reliability communication network. DSRC technology has some advantages over Wi-Fi (Wireless Fidelity) and GSM (Global System for Mobile Communications). It has a similar performance with WiMax (World Interoperability for Microwave Access) but the cost and the realization complexity is lower than WiMax. (Table1)

Tab.1 Comparison of DSRC and other communication technology [2]

	DSRC	Wi-Fi	GSM	WiMax
Delay	<50ms	Seconds	Seconds	/
Mobility	>60m/h	<5m/h	>60m/h	>60m/h
Data Transfer Rate	3-27 Mb/s	6-54 Mb/s	<2Mb/s	1-32 Mb/s
Communication Distance	<1000m	<100m	<10km	<15km
Communication Bandwidth	10MHz	20MHz	<3Mhz	<10MHz
Communication Band	5.86~ 5.925 GHz	2.4GHz, 5.2GHz	800MHz, 1.9GHz	2.5GHz
IEEE Standards	802.11P (WAVE)	802.11a	N/A	802.16e

There are proprietary frequency band in the application of DSRC in ITS which was allocated by Government (USA federal communication congress allocate 5.850~5.925GHz). The frequency recourse which is used in DSRC has been plotted to seven channels. The first channel is used for collision avoidance and vehicle-to-vehicle communication; the middle channel is used for control systems and control instructions to broadcast messages; the last channel is used for long distance and huge power communication; the other four channels which have been left are all service channel. (Figure1)

The standard communication protocol is developed based on IEEE802.11a. The bottom protocol and middle transition protocol has appeared. At last the IEEE802.11p protocol group has been developed. It is extended from IEEE 802.11 and the application from high speed transportation to standard speed 5.9KMHz (5.85-5.929KMHz) for V2V and V2I. [3]

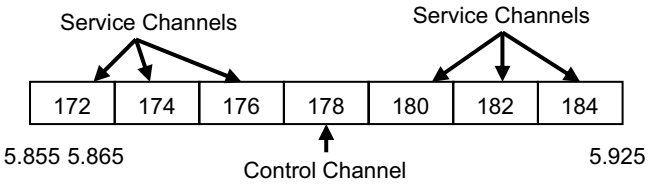


Fig.1 The allocation of DSRC communication channel

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3 The communication mechanism of DSRC

DSRC communication system commonly includes three segments: OBU (On Board Unit), RSU (Road Side Unit) and DSRC communication protocols. OBU is installed on the vehicle which runs in high speed. There are so many different equipments of OBU in the world. The communication mechanism and communication band are the key factors to distinguish these units. RSU is installed near the road shoulders. The characteristics of communication include transit frequency, transit power, communication interface and etc. DSRC protocols has three layers included physical layer, data link layer and application layer. [4]

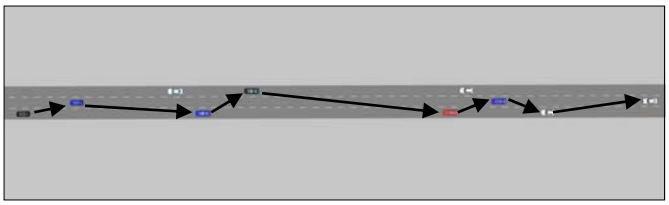
V2I is one kind of communication between mobile equipment and fasten equipment. The network model is based on single-Hoc communication. V2I was mainly used in ETC, media download and intelligent park system. V2V is one kind of communication between two mobile objects and its network model was based on multi-Hoc communication. The typical application of V2V is initiative avoidance, barrier detection and highway fleet management.

3.1 Vehicle-to-Vehicle communication mechanism

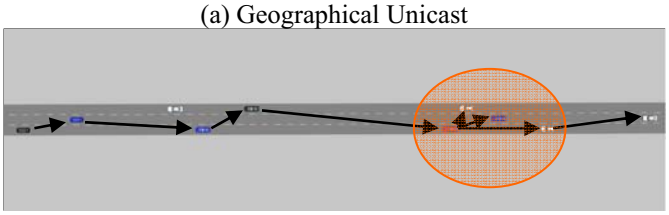
The Vehicle-to-Vehicle communication mechanism can be classified in three ways: geographical unicast, geographical broadcast and topologically-scoped broadcast. (Figure2) Geographical unicast provides packet delivery between two nodes via multiple wireless hops. When a node decides to send a unicast packet, it first determines the position of destination and sends the data packet to the node towards the destination. Geographical Broadcast distributes data packets by flooding, where nodes broadcast the packets if they are located in the geographical regions which are determined by the packets. This simple flooding scheme is enhanced with techniques based on packet numbering to alleviate the effects of broadcast storms that are problems in wireless ad hoc networks. Topologically-scoped broadcast provides rebroadcasting of a data packets from a source to all nodes in the n-hop neighbourhood. Single-hop broadcast is a specific case of TSB which is used to send periodic messages. [5]

3.2Vehicle-to-Infrastructure communication mechanism

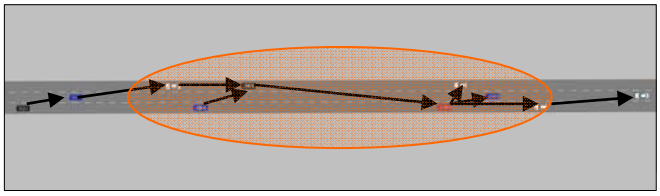
The onboard unit keep communication with infrastructure unit when the vehicle is running on the road. When the onboard unit leaves the radiofrequency range, it will disconnect the link of the current infrastructure unit and search for a new infrastructure unit. Once the new infrastructure unit has been found, the onboard unit will switch to the next infrastructure unit. There are 3 steps in switch progress: switch preparation, switch reregister and re-association stage. The infrastructure unit searching can be classified in two ways: active searching and passive searching. Active searching means the onboard unit sends requisition message to infrastructure actively and the infrastructure feedbacks the network topology information; Passive searching means the onboard unit receives the message from the infrastructure unit and get the topology information passively. [6]



(a) Geographical Unicast



(b) Geographical Broadcast



(c) Topologically-scoped Broadcast

Fig.2 Geocast Schemes

3.3 The network architecture of DSRC technology

The DSRC will be widely used in ITS in the future, so the network model will be different form different systems. This paper will discuss the DSRC network architecture of VII (Vehicle Infrastructure Integration) in USA (fig3). The integration Solution of the IPv6 protocol and geographical routing has been developed. This enables vehicles to use traditional applications based on the internet protocol, as well as to connect internet directly or even via multi-hop communication.

The government institutions and traffic management department sends the information which will be sent to drivers and passengers to the infrastructure unit via Ethernet network. Some enterprises can also send the message which will be broadcasted to public to the infrastructure unit via Ethernet. Then this message will be sent to the onboard unit and the drivers and passengers get this message selectively according to their specific need.

When the vehicle was driven to the toll station, the station can collect the toll from the driver via the communication between onboard unit on vehicle and infrastructure unit installed in the station. The system will search the customs' information from the database in information management center and record the toll information. If some accidents and incidents happen, the warning in formation will be broadcasted to the upstream drivers based on the vehicle-to-vehicle communication. The warning information can also be broadcasted from one region to another. [7]

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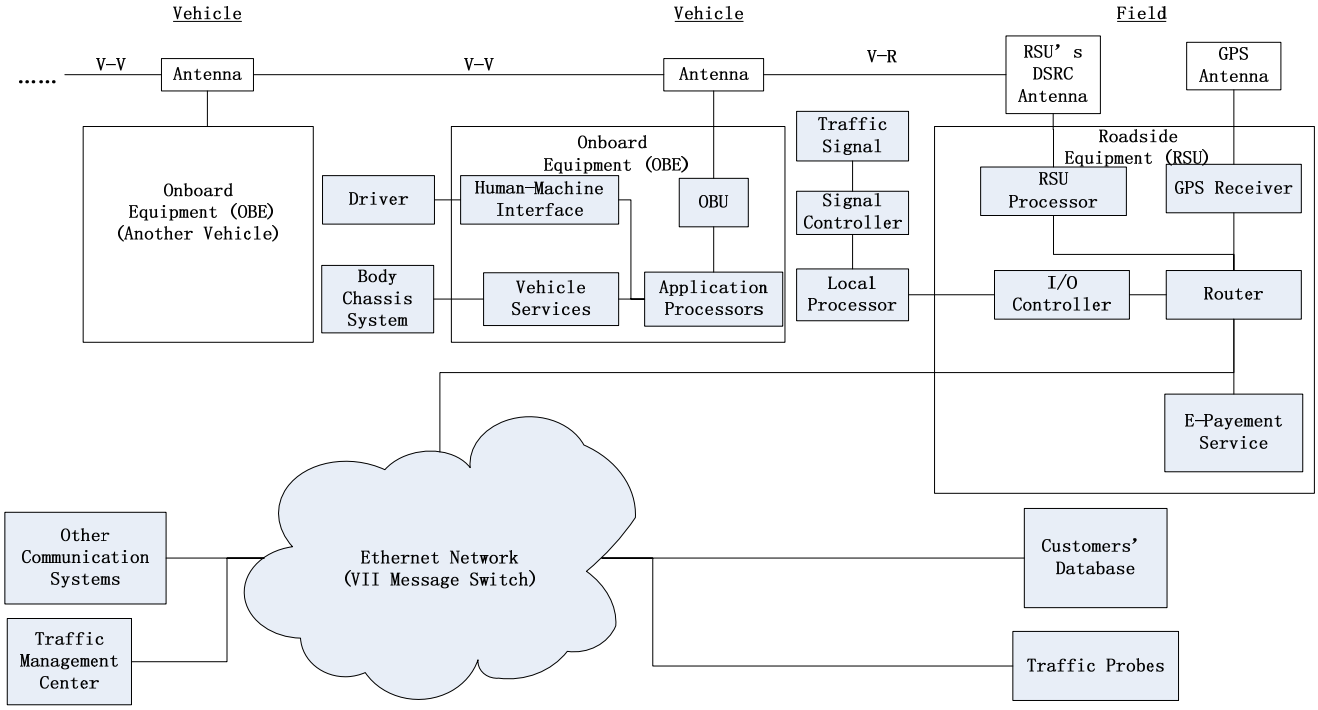


Fig.3 The DSRC network architecture in VII

4 The application of Vehicle-to-Vehicle communication

4.1 Traffic security

The Vehicle–to–Vehicle communication was mainly used in traffic security. The dangerous information can be sent from one vehicle to anther via the communication between the two vehicles. The probe installed on the vehicle can detect the barrier in front of the vehicle to avoid the collision accident. The probe can also detect the curve condition of the road to help the driver control the vehicle smoothly. [8] Figure4 shows a typical example. In the highway, the driver in the vehicle A finds his vehicle was broken down, he sent this message to the upstream vehicles. Occasionally there are a huge truck between vehicle A and vehicle B, the driver in vehicle B couldn't see the accident. But the onboard unit installed on the vehicle B could receive the message form vehicle A, the driver got the warning message timely and braked his vehicle in a safe distance.

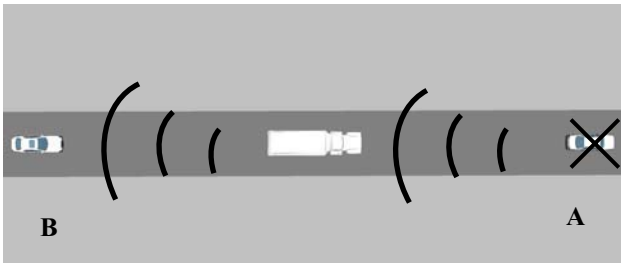


Fig.4 The initiative avoidance of the barrier

4.2 Intelligent traffic management

Intelligent traffic management mainly includes highway fleet management, emergency vehicle management, safe overtaking and etc. Once some emergences happen, the emergency vehicles could run in the green channels which are made by vehicle-to-vehicle communication. Figure 5 shows a typical example of highway fleet management: when the vehicle A and vehicle B run on the highway at high speed, the driver in vehicle C want to change the lane and join the fleet. Then the vehicle C sends a lane changing message to vehicle A and vehicle B, so that vehicle A and Vehicle B could maintain a large headway between them and vehicle C could change the lane safely.

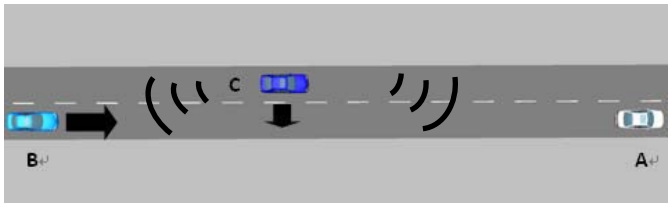


Fig.5 The highway fleet management

4.3 Real time traffic information transmission

Modern drivers often got the real time traffic information via traffic radio, Variable Message Signs (VMS) and mobile phone. DSRC-based communication technology can be offered a new way and used for real-time traffic information broadcasting in a specific region. [9] When a road traffic accident or traffic jam happened, the position and the time information determined by the GPS device installed on the surrounding vehicles can be transmitted to the upstream vehicles. Then the drivers in the upstream vehicles can decide

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to change the route to bypass the accident or the traffic jam spot.

5 The application of Vehicle-to-Infrastructure communication

5.1 Toll service

DSRC was widely used in the highway toll service. However, DSRC could be used in other for the multipurpose uses, including gas stations and parking lots. In these applications, the roadside infrastructure units should be installed in the proximity of the gas station or in the proximity of the ticketing and payment machine at the entrance and exit in the case of a pay-by-the-hour parking lot. The infrastructure unit could record toll and costumers' information and these information would be stored in the Customer Database. Sometimes these stations do not always have suitable location for installing additional equipment. If the antenna and control units are separated, the control unit can be easily installed in this place and the installation cost will increase a liite.

5.2 Vehicle management

Another Vehicle-to-Infrastructure service is used in vehicle management system, for controlling the entrance and exit of vehicles that have been registered in the system. When installed in a logistics warehouse or parking facilities of Residential community or factories, the system is used for the application as following: processing of the entry/exit of registered vehicles; improved security by preventing the trespassing of non-registered vehicles; archiving of electronic data on the history of entry/exit; improvement of entry/exit efficiency.

6 Summary and outlook

The characteristics of DSRC were suitable for the high speed movement. So DSRC technology has a broad application prospects in the intelligent transportation system and can be the trend of the development in the future.

However, there are also some security problems in the application of the DSRC technology in intelligent traffic system. Similar to other wireless communication technologies, the application should be protected from the attack of someone illegal. Once the security mechanism of communication was damaged, traffic accident would happen easily. The popularization of the communication devices can also affect the effect of the application of the DSRC technology. If there are only a few vehicles installed the on-board unit, the traffic problem could not be improved. In opposite, other vehicles without the on-board unit could be dangerous because of the disturbance of traffic flow caused by the vehicles with the on-board unit.

The investigation of the application of DSRC technology in ITS is an ideal start point for the development of other vehicle-to-vehicle and vehicle-to-infrastructure communication. A further step could be the solution for security problems of DSRC technology in ITS. For this purpose the improvement of the security mechanism should be discussed.

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